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# SIDELOBE SUPPRESSION FOR PHASED ARRAY AEROACOUSTIC MEASUREMENTS

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## ABSTRACT

Phased array test techniques are becoming increasingly popular in aeroacoustics. This is due, in part, to the use of optimized array designs (e.g., spiral shapes) which can operate over usefully wide frequency ranges with acceptable resolution and sidelobe rejection characteristics with practical numbers of microphones. Sometimes constraints such as limited space for microphone installation prevent the use of optimum array designs, raising the prospect of high sidelobe levels. Several modifications to classical beamforming which may reduce sidelobe levels for imperfect arrays (and possibly improve the performance of optimized arrays) are available. The effectiveness of these techniques is explored by applying three of them to four array designs using simulated wind tunnel data. The three beamforming techniques are robust adaptive beamforming, cross spectral matrix element weighting, and the CLEAN algorithm. The array designs are a cross, an array of four crosses, a filled square, and a spiral array.

## NOMENCLATURE

| | |
|---|---|
| N | = Number of microphones |
| $\bar{y}_n$ | = Location of microphone $n$, n=1...N |
| M | = Number of point sources |
| $\bar{x}_m$ | = location of source m, m = 1...M |
| $\bar{u}(t)$ | = Narrowband array data vector |
| $S_m(t)$ | = Narrowband source strength |
| $\bar{C}(\bar{x})$ | = Array steering vector for point $\bar{x}$ |
| $P(t)$ | = Scattering matrix associated with turbulence in boundary layer |
| $\bar{W}(t)$ | = Gaussian noise vector |
| $\langle\ \rangle$ | = Time average, also assumed to give expected value |
| A | = Cross spectral matrix |
| κ | = Normalization factor for $\bar{C}(\bar{x})$ |
| $r_{nm}$ | $= \lvert\bar{y}_n - \bar{x}_m\rvert$ |
| $\bar{d}_{nn'}$ | $= \bar{y}_n - \bar{y}_{n'}$ |
| $\rho_{nn'}$ | $= \lvert\bar{y}_n - \bar{y}_{n'}\rvert$ |
| G | = Decay constant for boundary/shear layer effect |
| $b_{bf}$ | = Beamform map function for classical beamforming |
| $b_{csmw}$ | = Beamform map function for beamforming with cross spectral matrix weighting |
| $b_{cl}$ | = Beamform map function for beamforming with the CLEAN algorithm |
| $b_o$ | = The highest level in a beamform map |
| $b_{psl}$ | = The peak sidelobe level in a beamform map |
| $R(n,n')$ | = Vector spacing redundancy index used in cross spectral matrix weighting |
| $v_{nn'}$ | = Weight for CSM element $(n,n')$ |
| B | = Result of weighting the elements of A |
| $\bar{w}$ | = Microphone weighting vector in minimum variance or adaptive beamforming, used like $\bar{C}$ in classical beamforming |
| α | = Loop gain parameter in CLEAN algorithm |

## INTRODUCTION

In classical beamforming, the phased array is essentially focused to each possible source location. When a source exists at the focusing point, a large response occurs. As the focus point is moved away from the source location, the output decreases. The array beamwidth is the size of the region for which the beamformer output is within 3 dB of the peak. Peaks may also occur at certain locations that are separate from a true source location. For regular, periodic, arrays, aliases with the same strength as the true source occur when $h > \frac{\lambda}{2}$, where h is microphone spacing and λ is the wavelength. The restriction on the acoustic frequency, $f$, required to avoid aliases is $f < c/(2h)$, where c is the speed of sound. This is a serious limitation of periodic arrays.

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Spurious peaks with lower levels than the real source, or sidelobes, occur for regular and irregular arrays. Two classification of sidelobes can be considered. One type is governed largely by the overall array shape, and occurs with uniformly sampled apertures as well as discretely sampled phased arrays. A familiar example of this type of sidelobe is the first outer ring that is 13 dB down from the central peak of the Airy disk of a circular aperture[1]. These sidelobes are commonly reduced by shading the array sensitivity near the perimeter. The second type of sidelobe, the subject of this paper, is governed by the discrete pattern of the microphones, or the array design. In an array with a regular microphone arrangement, certain vector spacings between pairs of microphones are repeated many times. This means that a plane wave incident from any direction will produce the same phase difference across each of the redundant microphone pairs (the set of pairs with the same vector spacing). If this spacing is greater than half of the wavelength, then there will be several distinct directions of incidence which can produce the same phase difference across these pairs.

The sidelobe rejection parameter, *SLR*, is defined here as the ratio of the peak level divided by the peak sidelobe level in dB, $SLR = 10 \log_{10}(b_0/b_{psl})$. If an array has $SLR = 10$ dB at a given frequency, and the real source gives an array output of 100 dB, then spurious peaks at 90 dB will occur at certain locations. This may be acceptable if the primary interest is in sources at or near the highest level; values lower that 10 dB below the highest can be ignored. Regular cross-shaped arrays have $SLR = 5$ dB at frequencies below $c/(2h)$ (and 0 dB for the aliases at higher frequency). Generally, this is not acceptable.

## TECHNIQUES FOR SIDELOBE REDUCTION

Array design along with beamform processing techniques determine the relative sidelobe levels. Three beamforming modifications for sidelobe reduction are considered: cross-spectral matrix element weighting, robust adaptive beamforming, and the CLEAN algorithm.

### Array design

Log spiral arrays were developed at Boeing (by Dougherty in 1989 and extended to the multi-arm case by James Underbrink[2] in 1994) to avoid aliases at all frequencies and offer useful sidelobe performance over a wide range of frequencies. Multi-arm spiral arrays of 64 elements can have $SLR > 7.5$ dB for all frequencies that have been examined[3].

Spiral arrays, and other low-sidelobe designs, are non-redundant; the relative vector spacings between the microphones are not repeated between different pairs[2]. Non-redundant design requires that the microphones be spread across the array aperture in a special pattern. Practical considerations frequently limit microphone placement in a way that prohibits a non-redundant array design. For example, microphones that are flush-mounted in the wall of a wind tunnel may be excluded from areas occupied by other instrumentation or structural elements of the wall. In another example, microphones positioned on the surface of a wind tunnel model are restricted to the geometry of the model itself. An aircraft planform may lend itself to a cross-array shape with one arm along the fuselage and the other arm on the wings. When large arrays are required for low frequency applications, cost and logistical problems may favor arrays with microphones along towers and other linear structures. In such cases, it is advantageous to use modified beamforming algorithms which reduce sidelobes.

### Cross-Spectral Matrix Element Weighting

The sidelobe performance of a cross array can be improved slightly by excluding from the beamforming process those elements of the cross-spectral matrix (CSM) which correspond to pairs of microphones in which both microphones belong to the same arm of the cross[4]. This is an example of the cross-spectral matrix element weighting approach; approximately half of the elements are weighted by 0. The first approach presented in this paper is a CSM element weighting scheme in which redundancies in the inter-element spacing are identified and corresponding matrix elements are de-weighted.

### Robust Adaptive Beamforming

The minimum variance algorithm is a beamforming technique which is derived by minimizing the array output subject to the condition that the gain is unity for a source at the steering location. Sidelobes are substantially reduced. The algorithm is adaptive

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because the coefficients with which the CSM elements are combined are adjusted to account for the particular CSM. The minimum variance algorithm is too unstable for aeroacoustic applications, but a modified form, robust adaptive beamforming, has been used[5]. Robust adaptive beamforming is the second technique explored here.

The CLEAN algorithm

A beamform map can be viewed as the result of convolving the source distribution function with the array point spread function. Sidelobes, which are spurious peaks in the point spread function, create clutter in the resulting map. If the point spread function is known, the source distribution can, in principle, be deconvolved from the beamform map. The CLEAN algorithm is a deconvolution technique which has been used in radio astronomy[6] and remote sensing[7]. Broadly, the approach is to identify the highest point in the beamform map, compute the point spread function for a source with that level at that location, subtract a fraction, α, of the point spread function from the map, and then iterate until the map has been reduced to zero. This is the inspiration for the third technique to be presented.

## BEAMFORMING ALGORITHMS

### Data model

Let $\bar{u}(t)$ be the complex $N$-vector of phased array microphone signals. The elements of $\bar{u}$, which result from application of a narrowband filter to the acoustic frequency signals, vary slowly in amplitude and phase. Suppose there are M point sources at locations $\bar{x}_m$ with complex narrowband strengths $S_m(t)$ for m = 1,...,M. For each source location, an array steering vector of N elements, $\bar{C}(\bar{x}_m)$, gives the amplitude and phase for undisturbed propagation from $\bar{x}_m$ to each of the microphones in the array. For a monopole source in free space with no flow:

$$\bar{C}(\bar{x}_m) = \kappa \frac{e^{(ikr_{nm})}}{r_{nm}}$$

where $r_{nm}$ is the propagation distance between source location $\bar{x}_m$ and microphone location $n$. κ is a normalization constant such that $\bar{C}^\dagger \bar{C} = 1$. When flow is present, the argument of the exponential and the denominator are modified to account for the source-receiver travel time. Propagation through a turbulent boundary layer or shear layer to microphone $n$ is assumed to have an additional, statistical effect described by a random process P(t). The array data model becomes:

$$\bar{u}(t) = P(t)\sum_{m=1}^{M} \bar{C}(\bar{x}_m) S_m(t) + \bar{W}(t) \qquad (1)$$

where

$$P(t) = Diag\left[P_n(t)\right], \quad n = 1,\ldots,N,$$

and the noise vector, $\bar{W}(t)$, is assumed to be independent of $P(t)$ and $S_m(t)$, m = 1, M.

The array cross-spectral matrix is given by:

$$A = \left\langle \bar{u}\bar{u}^\dagger \right\rangle \qquad (2)$$

where $\langle\ \rangle$ is a time average (over several seconds), which is also assumed to give the expected value. Using (1), the array CSM becomes

$$\begin{aligned} A &= \left\langle P \sum_{m,m'} \bar{C}(\bar{x}_m)\bar{C}^\dagger(\bar{x}_{m'}) P^\dagger S_m S_{m'}^* \right\rangle + \left\langle \bar{W}\bar{W}^\dagger \right\rangle \\ &= \sum_{m,m'} \left\langle P\bar{C}(\bar{x}_m)\bar{C}^\dagger(\bar{x}_{m'})P^\dagger \right\rangle \left\langle S_m S_{m'}^* \right\rangle + \left\langle \bar{W}\bar{W}^\dagger \right\rangle \end{aligned} \qquad (3)$$

where the second form assumes that $P(t)$ and the source strengths are statistically independent. The term $\left\langle S_m S_{m'}^* \right\rangle$ can be described as a source cross-spectral matrix element. In this model, the M sources may be independent, coherent with certain phase relationships, or partially coherent. Element $(n, n')$ of $A$ is given by:

$$\begin{aligned} A_{nn'} &= \left\langle P_n P_{n'}^* \right\rangle \sum_{m,m'} C_n(\bar{x}_m) C_{n'}^*(\bar{x}_{m'}) \left\langle S_m S_{m'}^* \right\rangle \\ &\quad + \left\langle W_n(t) W_{n'}(t) \right\rangle \end{aligned} \qquad (4)$$

The effect of $P(t)$ is to weight each CSM element $A_{nn'}$ by $\left\langle P_n P_{n'}^* \right\rangle$. Experience in aeroacoustic testing indicates that this factor is a decreasing function of $\frac{\rho_{nn'}}{\lambda}$, where $\rho_{nn'}$ is the microphone spacing and λ is

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the acoustic wavelength. The assumption for this study is:

$$\left\langle P_n P_{n'}^* \right\rangle = e^{-\frac{\rho_{nn'}}{G\lambda}}$$

where G is a constant depending on the test setup and the boundary/shear layer details.

Classical Beamforming

The expression for the beamforming source strength map is:

$$b_{bf}(\bar{x}) = \bar{C}^{\dagger}(\bar{x}) A C(\bar{x}) \tag{5}$$

for a single source at $\bar{x}_o$, and $P = I$, this reduces to:

$$b_{bf}(\bar{x}) = \left\langle S_o S_o^* \right\rangle \left| \bar{C}^{\dagger}(\bar{x}) \bar{C}(\bar{x}_o) \right|^2 \tag{6}$$

which is the average-square source strength times the array pattern. Any flaws in the array pattern will be translated to the final results if equation (5) is used for array processing. The main effect of finite G (giving non-identity $P$) is to reduce the level of the map. Noise terms tend to cancel in the beamforming sum, giving a robust method.

CSM Element Weighting

Let the array microphone locations vectors be denoted $\bar{y}_n$ for n = 1,...,N and let the vector spacing between microphones $n$ and $n'$ be denoted $\bar{d}_{nn'} = \bar{y}_n - \bar{y}_{n'}$. In considering microphone pairs, such as $(n,n')$, it will be assumed that $n < n'$, since $(n',n)$ refers to the same pair. If $(n,n')$ and $(l,l')$ denote two microphone pairs, then the length of the difference between the interelement spacings is :

$$D\left[(n,n')(l,l')\right] = \min\left\{\left|\bar{d}_{nn'} - \bar{d}_{ll'}\right|, \left|\bar{d}_{nn'} + \bar{d}_{ll'}\right|\right\}.$$

(The second term in braces accounts for the possibility that $\bar{d}_{nn'}$ and $\bar{d}_{ll'}$ may be oppositely directed.) If this is substantially less than the acoustic wavelength, λ, then $(n,n')$ and $(l,l')$ are redundant pairs, since their vector spacings are effectively identical. Let a redundancy index, $R(n,n')$ for the given pair, $(n,n')$, be defined as the number of other pairs that have the same, or nearly the same, vector spacing as $(n,n')$. An estimate of $R(n,n')$ can be computed

$$R(n,n') = \sum_{\substack{(l,l') \neq (n,n') \\ l<l'}} f\left(k_o D\left[(n,n'),(l,l')\right]\right) \tag{7}$$

where,

$$f(k_o D) = e^{-(2k_o D)^2} \tag{8}$$

and

$$k_o = \tfrac{2\pi}{\lambda}$$

Here the proximity function, $f(k_o D)$, has been constructed to approach unity for $k_o D << 1$ and to decrease rapidly for $k_o D > 0.5$.

Suppose, for example, that $R(n,n') = 5$ for a certain pair, $(n,n')$. To prevent this circumstance from creating sidelobes, it is desirable to diminish the effect of $(n,n')$, as well as the other 5 redundant pairs, by a factor of 1/6. The particular spacing, $\bar{d}_{nn'}$, will then have the same influence in the beamforming result as a non-redundant pair with spacing $\bar{d}_{nn'}$ would have. This diminution can be achieved by scaling each element of the array CSM matrix by the factor

$$v_{nn'} = \frac{1}{1 - R(n,n')} \tag{9}$$

(and subsequently renormalizing the entire matrix by dividing by the average value of $v_{nn'}$ in order to approximately retain the same gain). For array CSM elements corresponding to non-redundant spacings, $R(n,n') = 0$, so the weighting has no effect (apart from the final renormalization). In the example case of a cluster of 6 points in the coarray (a scatter plot of the vector spacings), $R$ will be 5 for each of the pairs, and each of the corresponding matrix elements will be multiplied by 1/6. The beamforming expression, equation (5), which can be viewed as a sum of the cross spectral matrix elements (adjusted by the appropriate phase shifts to steer the beam to $\bar{x}$) will include redundant pairs with the proper weighting to reduce sidelobes.

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The CSM element weighting approach to sidelobe reduction can be summarized by:

1. Compute the element factor $v_{nn'}$ for each microphone pair $(n, n')$

2. Form the weighted and normalized array CSM, B, with elements

$$B_{nn'} = \frac{\sum_{ll'} |A_{ll'}|}{\sum_{ll'} |v_{ll'} A_{ll'}|} v_{nn'} A_{nn'} \qquad (10)$$

3. Compute the beamform map according to

$$b_{csmw}(\bar{x}) = \bar{C}^{\dagger}(\bar{x}) B \bar{C}(\bar{x}) \qquad (11)$$

Robust Adaptive Beamforming

As presented in reference [5], the minimum variance and robust adaptive beamforming algorithms use the beamforming expression, (5), with the steering vector replaced with an alternate element weighting vector. For minimum variance, $\bar{w}$ is given by

$$\bar{w} = \frac{A^{-1}\bar{C}}{\bar{C}^{\dagger} A^{-1} \bar{C}} \qquad (12)$$

This element weighting gives a beamforming map with a minimum value at each point, $\bar{x}$, subject to the constraint that the array has unity gain (defined by $\bar{w}^{\dagger}\bar{C} = 1$). The array sidelobes and beamwidth are automatically minimized in a way that accounts for the particular cross spectral matrix. The matrix $A$ is usually ill-conditioned or singular so (12) can not be applied. This problem is overcome by adding a small diagonal matrix to $A$ before inverting it, giving the weighting vector for robust adaptive beamforming:

$$\bar{w} = \frac{(A + \varepsilon I)^{-1}\bar{C}}{\bar{C}^{\dagger}(A + \varepsilon I)^{-1}\bar{C}} \qquad (13)$$

The value of ε is adjusted for optimum performance; the limits of 0 and infinity reduce to minimum variance and classical beamforming. (A complex justification for ε is given in the literature.)

The CLEAN Algorithm

In the CLEAN algorithm[6], classical beamforming is first performed, at a grid of points, $\bar{x}$, giving what is known as the "dirty" map (the true source distribution modified by the array measurement and beamforming process). It is reasonable to expect the peak level of the dirty map to correspond to the peak source location, since the amplitude of the sidelobes are below the peak amplitude. A "clean" map, $b_{cl}$, is constructed iteratively from the dirty map. A residual map, $b_{res}$, is initialized to the value of the dirty map, and $b_{cl}$ is initialized to zero. Steps 1 through 4, below, are repeated until the residual map has been reduced sufficiently such that no valid peaks can be found:

1. Identify the largest and smallest values in $b_{res}$. The largest value, at map point $\bar{x}_p$, is a valid peak if the sum of the largest and smallest values is positive (otherwise, $b_{res}$ has been reduced sufficiently that further iterations are not likely to improve $b_{cl}$). Let the peak level be $b_p$.
2. Add $\alpha b_p$ to $b_{cl}$ at the grid point $\bar{x}_p$. The loop gain parameter, α, is chosen typically in the range of 0.1 to 0.25.
3. Compute the array pattern for a source at $\bar{x}_p$ with strength $\alpha b_p : \alpha b_p \left| \bar{C}^{\dagger}(\bar{x}) \bar{C}(\bar{x}_p) \right|^2$ and subtract this from $b_{res}$.
4. Repeat steps 1-3.

Ideally, the clean map should converge to a set of source strengths which, if used in equations (3) and (5) (with $P = I$, and diagonal $\langle S_m S_{m'}^* \rangle$) would reproduce the dirty map. Questions of how $b_{cl}$ is related to the true source distribution are best studied numerically, especially if sources are mutually correlated or measurement errors such as nonidentity $P$ are involved.

## RESULTS

Four arrays were used for the numerical simulations described, a cross array, a filled square array, a spiral array and a multiple cross array as shown in Figure 1. The cross array was 1.07m wide by 1.07m high and was composed of 65 elements. The filled square array was 1.07m wide by 1.07m high and composed of 64 elements. The spiral array had an outer diameter of

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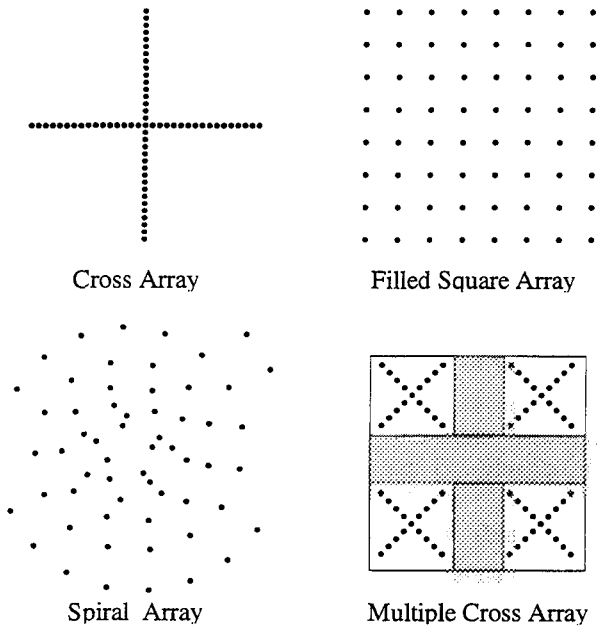


Figure 1. Four arrays used in numerical simulations.

1.07m and was composed of 64 elements. The multiple cross array shows a possible array design that is constrained by two structural beams which reduce the area where microphones may be placed. In this design, it is presumed that multiple windows must be used because no single window is large enough to give the desired resolution. In this multiple cross array, each cross is 0.4m by 0.4m. The overall dimension is 1m by 1m and the array has 68 microphones.

Synthetic data were created at frequencies from 1-40 kHz by assuming a monopole source 2.2m from each array center. An attempt was made to reproduce the features of aeroacoustic data acquired from an array flush-mounted in the wall of a wind tunnel running at a Mach number of 0.25. The CSM decorrelation factors, $\langle P_n P_{n'}^* \rangle$, were determined with a spatial decay constant of G = 25. Gaussian random noise, $\langle \bar{W}\bar{W}^\dagger \rangle$, was added to each CSM with a signal to noise ratio, the ratio of the average magnitude of noiseless CSM elements to the standard deviation of the Gaussian noise elements, of 3 dB.

The four beamforming algorithms were applied using a planar 4.5m by 4.5m grid of $1.6 \times 10^5$ points containing the true source point at the center of the grid. The factor ε for robust adaptive beamforming was set at 20, the smallest value that avoided serious instability problems at high frequency.

The resulting beamforming maps were analyzed to determine the peak levels, beamwidths and sidelobe rejection factors. In all cases, the highest level in the map occurred at the grid point corresponding to the true source location. The peak level was below the true source level by an amount that increased from 0 dB at 1 kHz to 6-8 dB at 40 kHz. This level decrease, which is also seen in real test results, is caused by the microphone decorrelation represented by the factor $\langle P_n P_{n'}^* \rangle$.

The beamwidth and sidelobe evaluation of the CLEAN results was problematic. The central peak stands out from the values at all other grid points by at least 12 dB at all frequencies for each of the arrays (with the exception of the multiple cross array above 28 kHz). Strictly speaking, the 3 dB beamwidth is zero, and the levels for all grid points other than the source location should be considered to be associated with sidelobes. However, nonzero, non-peak levels in the maps at low frequency tend to be tightly clustered around the peak; they are not likely to be mistaken as additional sources. It is therefore not appropriate to identify these artifacts as either sidelobes or part of the peak. The difficulty was resolved by applying a nine-point smoothing operator to the map before evaluating the peak widths and sidelobe levels. This broadened the CLEAN peaks to a full width of two grid spacings, but prevented points very close to the peak from being identified as sidelobes, in most cases.

The beamwidth and sidelobe results are given in Figures 2-5. The sidelobe rejection is artificially limited to 30 dB by a file formatting issue. This only affects the CLEAN results.

## CONCLUSIONS

### Decorrelation effects

The falloff of peak level with frequency that is seen in real data is consistent with the decorrelation model presented here. This should be considered in array design and interpretation of results.

### Comparison of algorithms

In terms of beamwidth performance, the algorithms ordered from best to worst are: CLEAN, robust

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adaptive beamforming, CSM weighting, and classical beamforming. This holds for all of the arrays and frequencies considered. (The fact that the beamwidth for CLEAN appears to surpass the others slightly at high frequency is an artifact of the smoothing process described.)

The ordering of the algorithms for sidelobe performance is the same as the order for beamwidth.

Array-algorithm combinations

Array design and algorithm selection should be done together. The combination of the spiral array with classical beamforming produces consistent sidelobe results. Only CLEAN produces substantial improvement with this array, although the robust adaptive method has some benefit at the lower frequencies.

The filled square array is useful over a wide frequency range only with CLEAN.

The multiple cross array has poor but consistent sidelobe performance with classical beamforming. Each of the other algorithms improves the sidelobes, but CLEAN appears to be required for useful results.

All of the methods improve the sidelobes of the cross array.

## ACKNOWLEDGEMENTS

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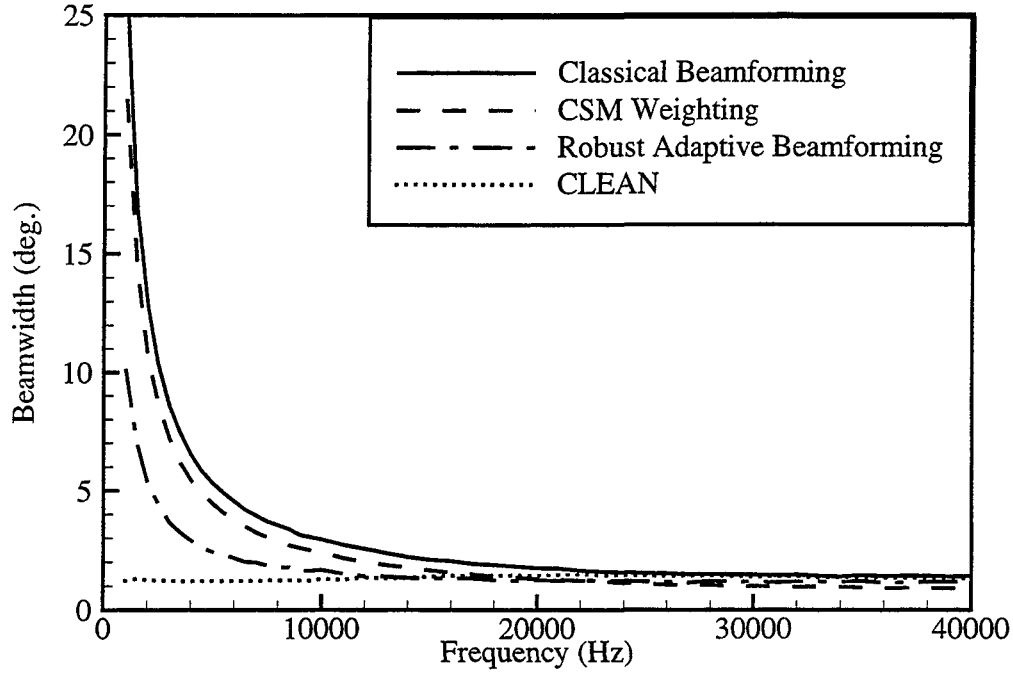


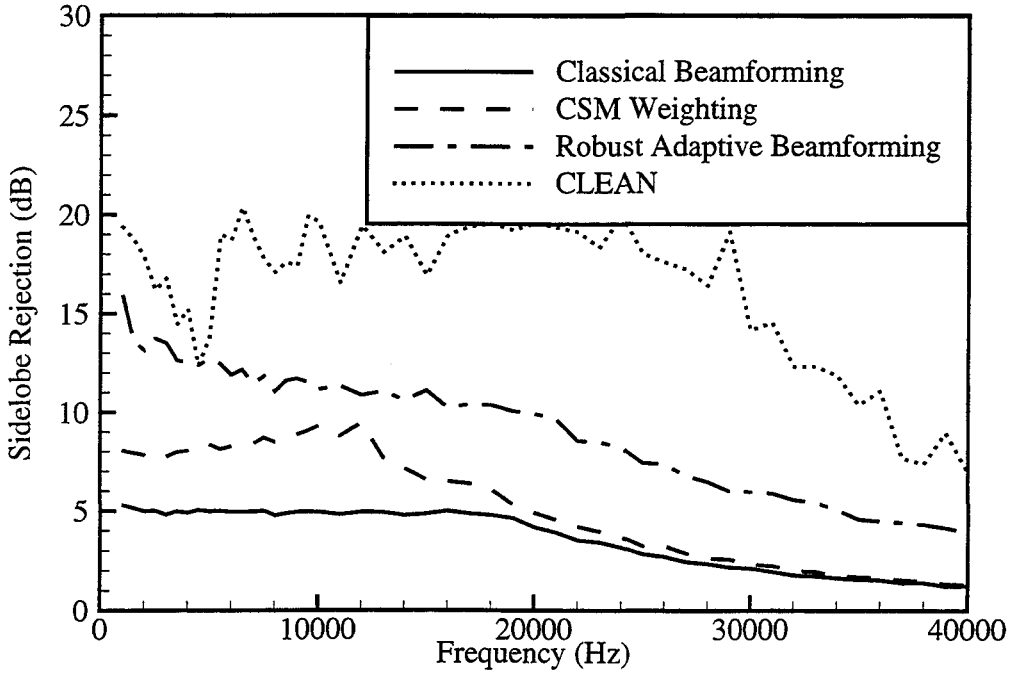


Figure 2. Results from the simulations using the Cross Array.

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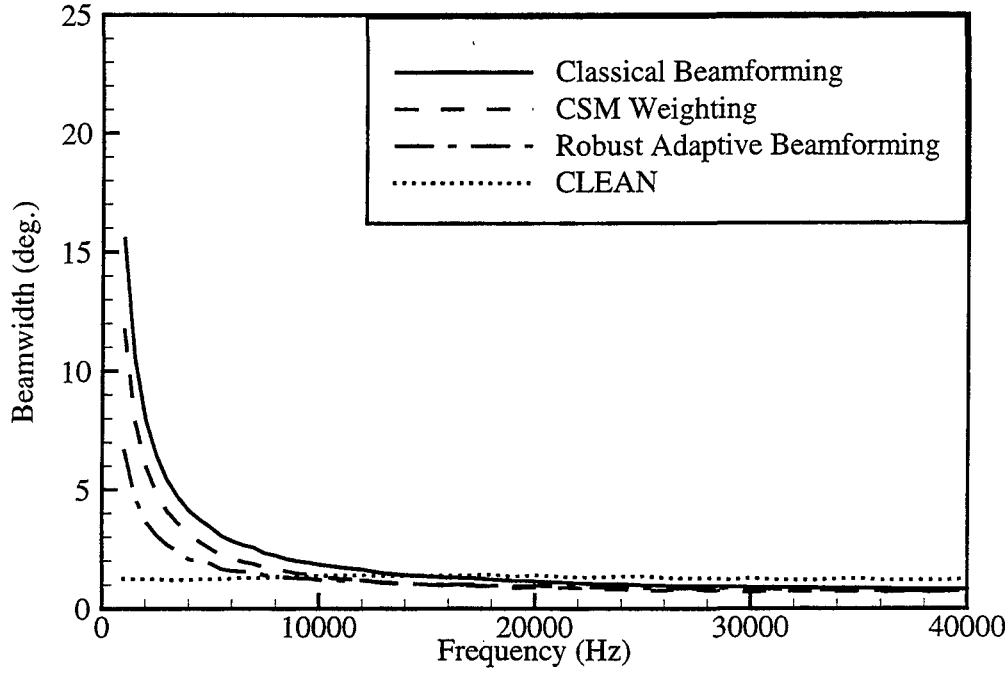


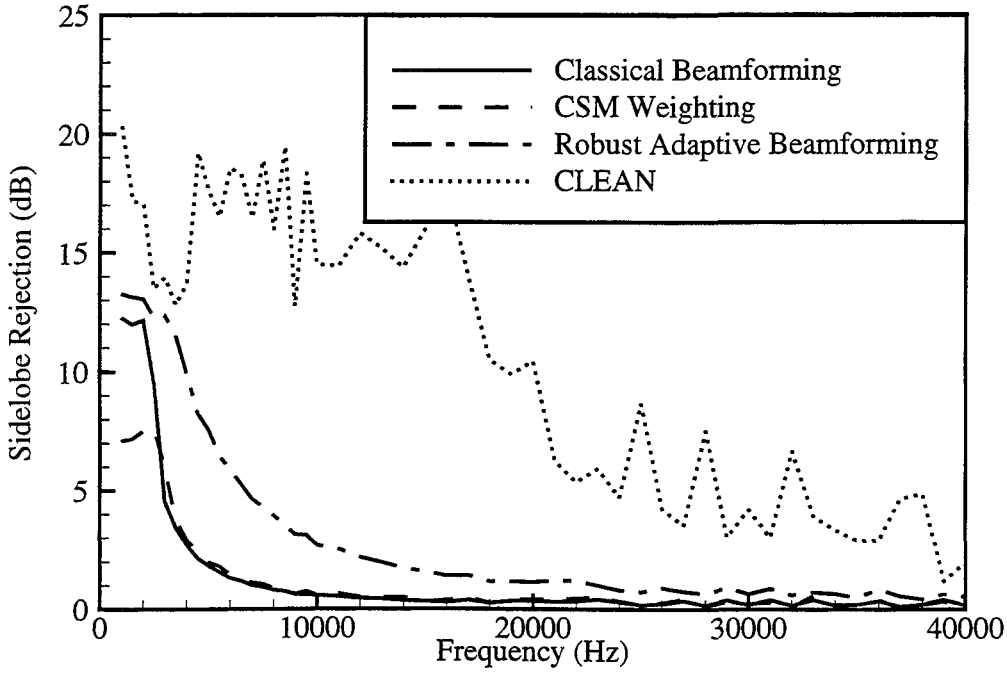


Figure 3. Results from the simulations using the Filled Square Array.

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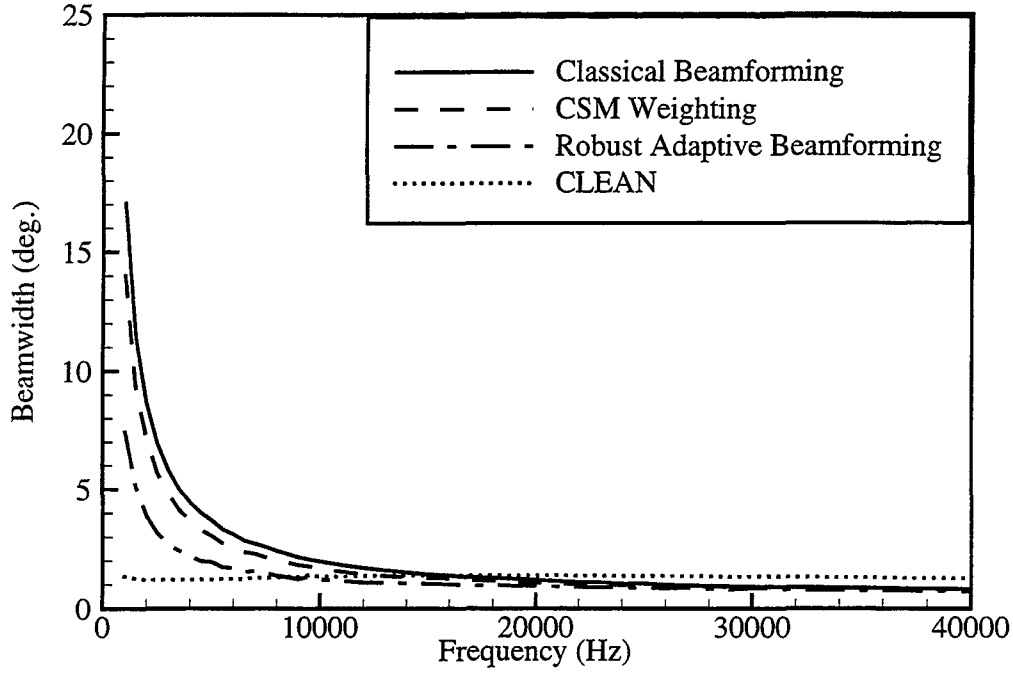


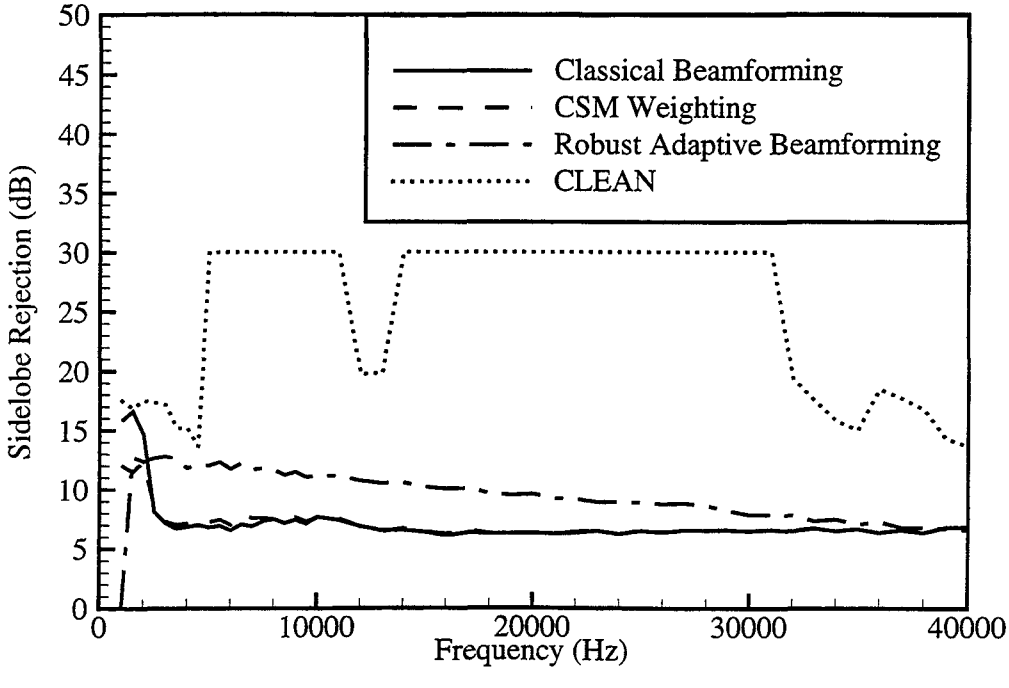


Figure 4. Results from the simulations using the Spiral Array.

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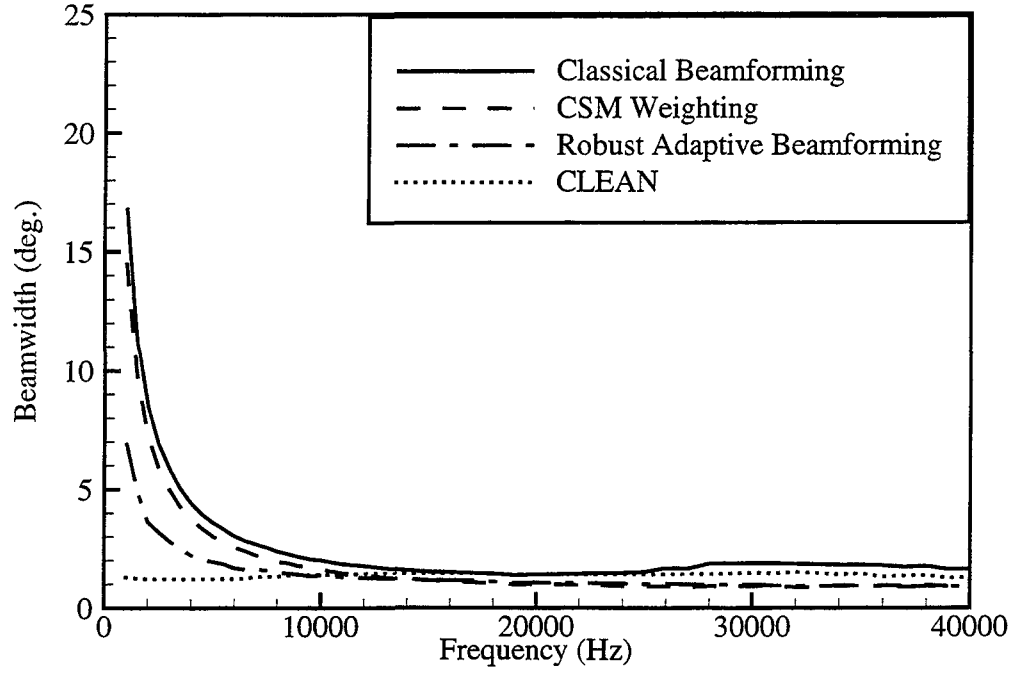


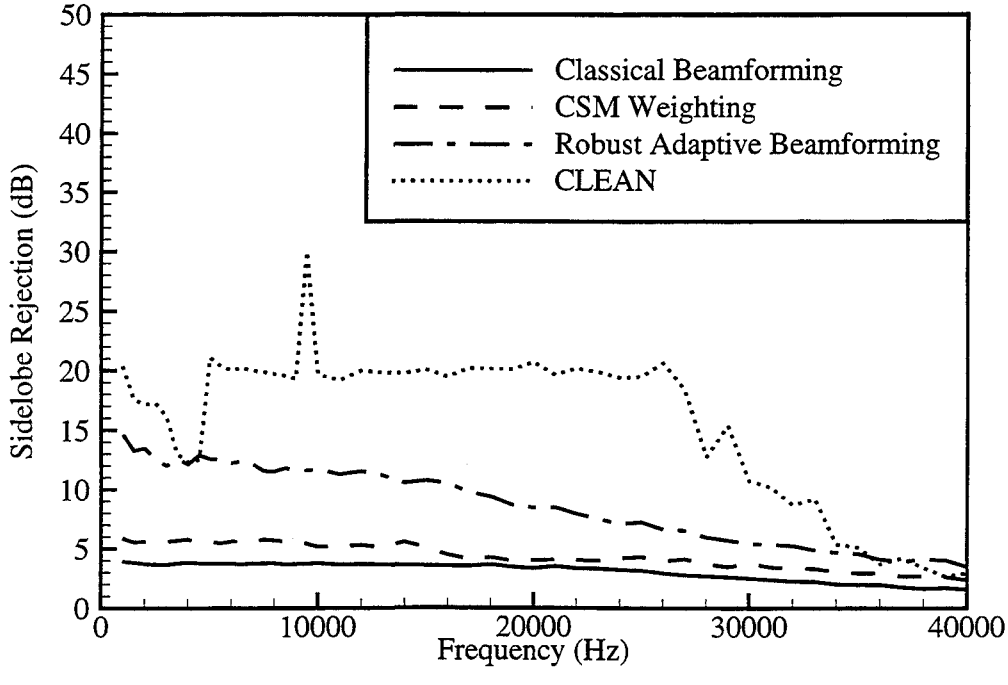


Figure 5. Results from the simulations using the Multiple-Cross Array.